

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)

Master of Business Administration (MBA)

Post Graduate Diploma in Management (PGDM)

Semester –I University Examination – March 2018

MB701.3 Quantitative Analysis for Management – 1 (New)

MB701.2 Quantitative Analysis for Management – 1 (Old)

27.03.2018, Tuesday

Time: 01:30 PM to 04:30 PM

Maximum Marks: 70

Instructions:

1. This paper contains six questions.
2. All questions are compulsory.
3. Each of the first five questions will carry 10 marks and the last question will carry 20.
4. Tables shall be provided by the institute / university.

- Q-1 (a)** Of the 1125 students studying in a college during a year, 720 were SC/ST, 628 were boys, and 440 were science students; the number of SC/ST boys was 392, that of boys studying in Science 205, and that of SC/ST students studying science 262; finally the number of science students among the SC/ST boys was 148. You are required to present the above information in a suitable tabular form. **05**
- (b)** The number of cars sold by each of the 10 car dealers during a particular month, arranged in ascending order, is 12, 14, 17, 20, 20, 20, 22, 22, 24, 25. Considering this scale to be the statistical population of interest, determine the mean, median and mode for the number of cars sold. Which value calculated above best describes the 'typical' sales volume per dealer? **05**
- Q-2 (a)** Assume that a factory has two machines. Past records show that machine 1 produces 30 per cent of the items of output and machine 2 produces 70 per cent of the items. Further, 5 per cent of the items produced by machine 1 were defective and only 1 per cent produced by machine 2 were defective. If a defective item is drawn at random what is the probability that the defective item was produced by machine 1 or machine 2? **05**
- (b)** A new automated production process has had an average of 1.5 breakdowns per day. Because of the cost associated with a breakdown, management is concerned about the possibility of having three or more breakdowns during a day. Assume that breakdowns occur randomly, that the probability of a breakdown is the same for any two time intervals of equal length, and that breakdowns in one period are independent of breakdowns in other periods. What is the probability of having three or more breakdowns during a day? **05**
- Q-3** A diameter of a component produced on a semi-automatic machine is known to be distributed normally with a mean of 10 mm and a standard deviation of 0.1 mm. If a random sample of size 5 is picked up, what is the probability that the sample mean will be between 9.95 mm and 10.05 mm? **10**
- Q-4** Individual filing of income tax returns prior to 30 March had an average refund of Rs. 1200. Consider the population of 'last minute' filers who file their returns during the last week of March. For a random sample of 400 individuals who filed a return between 25 and 31 March, the sample mean refund was Rs. 1054 and the sample standard deviation was Rs. 1600. Using 5 per cent of level of significance, test the belief that the individuals who wait until the last week of March to file their returns to get a higher refund than early the filers. **10**

- Q-5** The following data give the ages and blood pressure of 10 women. **10**

Age	56	42	36	47	49	42	60	72	63	55
Blood Pressure	147	125	118	128	145	140	140	155	149	150

When tested regression on MS – Excel; following results were generated:

Regression Statistics

Multiple R	0.892
R Square	0.795
Adjusted R Square	0.769
Standard Error	6.607
Observations	10.000

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	1354.91	1354.91	31.04	0.00
Residual	8	349.19	43.65		
Total	9	1704.10			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	83.76	10.61	7.90	0.00	59.29	108.22	59.29	108.22
Age	1.11	0.20	5.57	0.00	0.65	1.57	0.65	1.57

- Use methods of descriptive statistics to summarize the data. Comment on the findings. Prepare a managerial report.
- Develop a regression model to predict the impact of Age on Blood Pressure.
- Discuss the need for other independent variables that could be added to the model, what additional variables might be useful.
- Estimate the blood pressure of a woman whose age is 45 years.

Q-6

Two hundred randomly selected adults were asked whether TV shows as a whole are primarily entertaining, educational, or a waste of time (only one answer could be chosen). The respondents were categorized by gender. Their responses are given in the following table: **20**

Gender	Opinion			
	Entertaining	Educational	Waste of Time	Total
Female	52	28	30	110
Male	28	12	50	90
Total	80	40	80	200

Is this evidence convincing that there is a relationship between gender and opinion in the population interest?

Use **chi-square test** of homogeneity to draw inferences from the data. Use $\alpha=0.05$